

SYSTEMATICALLY ADAPTING PROCESSABILITY AND FLAME RETARDANCY OF EPOXY GLASS FIBER REINFORCED COMPOSITES FOR RAILWAY APPLICATIONS

Sruthi Sunder^{*1}, Leonard Hieke^{*1}, Rodrigo Q. Albuquerque¹ and Holger Ruckdäschel¹

¹Department of Polymer Engineering, University of Bayreuth
Universitätstrasse 30, 95447, Bayreuth, Germany

Email: sruthi.sunder@uni-bayreuth.de, web page: www.polymer-engineering.de

* These authors contributed equally to the paper

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ABSTRACT

Epoxy glass fiber composites offer significant weight reductions for railway applications, reducing fuel consumption by 12–14%. However, the intrinsic flammability of the epoxy matrix poses challenges in achieving HL-2 fire classification under DIN EN 45545. This study addresses this limitation by systematically identifying and optimizing phosphorus-based flame retardants (FRs) compatible with vacuum infusion. An initial pool of 14 FR additives, including both solid (e.g., ammonium polyphosphate variants) and liquid (e.g., triethyl phosphate esters (TEP)) was screened in a Bisphenol F epoxy resin preloaded with 10 wt% resorcinol bis(diphenyl phosphate) (RDP). To reduce experimental effort, a batch active learning (B-AL) approach using Gaussian process regression (GPR) was applied to select 18 formulations for thermal and rheological characterization. Simultaneous thermal analysis (STA) parameters were used as surrogates for fire behavior in cone calorimetry, while pot life and storage stability were assessed to ensure processability. Surrogate models trained on STA and processability data achieved moderate predictive performance ($R^2 = 0.4\text{--}0.6$). Permutation importance was used to rank the FRs, identifying silane-coated APP (APPIII), melamine-coated APP (APPIV), TEP, and DOPOI as the most influential additives. These FRs were used in a secondary design space to generate 16 randomized formulations for transfer into woven glass fiber composites. Cone calorimeter tests conducted on top- and bottom-face exposures revealed increased filtration effects with higher solid FR content. The best balance between reduction in maximum average rate of heat emission (MARHE) from 155 kW/m² in the reference composite to 110 kW/m² and standard deviation in the MARHE values were found for the resin formulation containing 9% w/w of the liquid TEP with 1% APP and 10% RDP. Further optimization is required to satisfy railway norms. This machine learning-guided framework demonstrates a scalable route to optimize FR combinations that balance fire safety and processability.

1 INTRODUCTION

Lightweight fiber-reinforced composites are vital for the railway industry, offering corresponding fuel-efficiency improvements of 12–14 % which is a compelling driver for sustainability and reduced operating costs. However, these systems, particularly epoxy (EP)–glass fiber reinforced composites (GFRCs), pose significant fire-safety challenges due to the intrinsic flammability of the EP matrix, which fails critical safety thresholds required for passenger rail applications. Materials used in railway applications must comply with DIN EN 45545-2 [1], which outlines various parameters, most notably maximum average rate of heat emission (MARHE). For Hazard Level HL-2 compliance, the MARHE limit is 90 kW/m², while more stringent HL-3 applications demand values ≤ 60 kW/m². Meeting EN 45545-2 using phosphorus-based flame retardants (FRs) is one approach since they act synergistically in both the gas phase—by releasing radicals that suppress flame propagation—and the condensed phase, by promoting char formation to shield the polymer. For instance, aluminium diethyl phosphinate is a well-established flame retardant for epoxies operating via this dual mechanism [2–5]. Combinatorial systems like ammonium polyphosphate (APP) or intumescent blends have demonstrated

the potential to reduce peak heat release rate (pHRR) [6–9], and thus, MARHE values [10]. The relatively low viscosity (300–2000 mPa.s) required for vacuum infusion or resin transfer molding (RTM) processing, combined with the need to avoid filtration effects ^[11–13], constrains FR selection to additives that maintain processability.

While full-scale cone calorimetry (EN ISO 5660-1[14]) remains the gold standard for MARHE measurements, proxy screening via simultaneous thermal analysis (STA) using parameters like thermal decomposition onset and char yield can be indicative of fire performance ahead of composite fabrication [15]. For example (Figure 1), in differential scanning calorimetry (DSC), a left-shift in the heat energy of the first peak (Q1), or a decrease in intensity of the heat energy of the second peak (Q2) correspond to a decrease of pHRR and thus, MARHE values. Similarly, the adjusted heat enthalpy of the first thermal event (ΔH_{1adj}) in thermogravimetric analysis (TGA) also represents a decrease in the pHRR. These correlations were generated using logistic regression combined with a least absolute shrinkage and selection operator (LASSO) model for a Bisphenol F Diglycidyl Ether (DGEBF)-diethyl toluene diamine system by Binder et al. [15]. Although modified contexts may result in modified correlation strengths between STA and cone parameters, it is assumed that these correlations can be exploited to screen FR additives for transfer to composites.

Comprehensive experimental screening becomes resource-intensive given the combinatorial nature of FR systems. Krebs [16] investigated how to boost the flame-retardant performance of a high-temperature epoxy resin for both aerospace and railway applications. Screening synergistic mixtures of ALPi with zinc stannate (ZnSt), and two inorganic silicate additives, guided by Bayesian optimization (BO) via Gaussian process regression (GPR) helped simultaneously minimize a hybrid target with total smoke production (TSP) and MARHE. When transferred to GFRCs via prepreps, the BO-optimized formulation nearly satisfied a HL-2 classification. Modelling with both linear and nonlinear regressors showed that a Random Forest algorithm best captured the complex, nonlinear relationships between FR content and performance. Storage stability estimations over months for iterative machine learning (ML) experiments are time consuming; accelerated aging of resin formulations at temperatures $> 25^{\circ}\text{C}$ can be used to estimate storage stability based on differences in the initial glass transition temperature T_{g0} over shorter time periods [17].

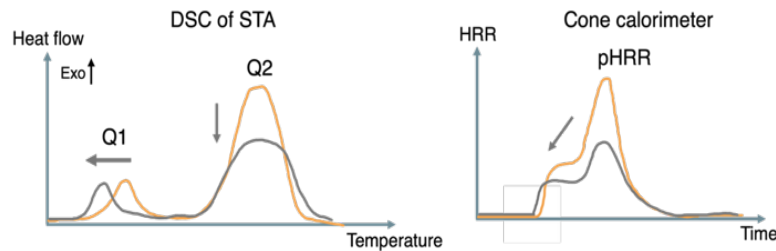


Figure 1: Correlations between heat flow in DSC measurements and pHRR measurements

The use of active learning (AL) using batch experiments (B-AL) with GPR also enables efficient exploration across the formulation space beyond BO [18–20]. This study builds upon these foundations by screening 14 phosphorus-based FR additives (gas- and condensed-phase) in a DGEBF resin containing 10 % w/w resorcinol bis(diphenyl phosphate)) (RDP). Processability via shear viscosity, and storage tests, targeting vacuum infusion compatibility. STA-informed B-AL/GPR models are used to gather data for training surrogate models for FR selection. Selected FRs based on feature importance estimations are transferred into 100 g/m² glass fiber composites via vacuum infusion across 16 randomized design trials, guided by BO. This approach integrates fire safety, processing constraints, and machine-learning-based surrogate modelling to enable accelerated material development in compliance with EN 45545 for weight-efficient and safe composite components.

2 MATERIALS AND METHODS

An overview of the methods used in this investigation is shown in Figure 2. B-AL is used as the exploratory phase to eventually reduce the number of features (FRs) for future exploitative Bayesian optimisation experiments.

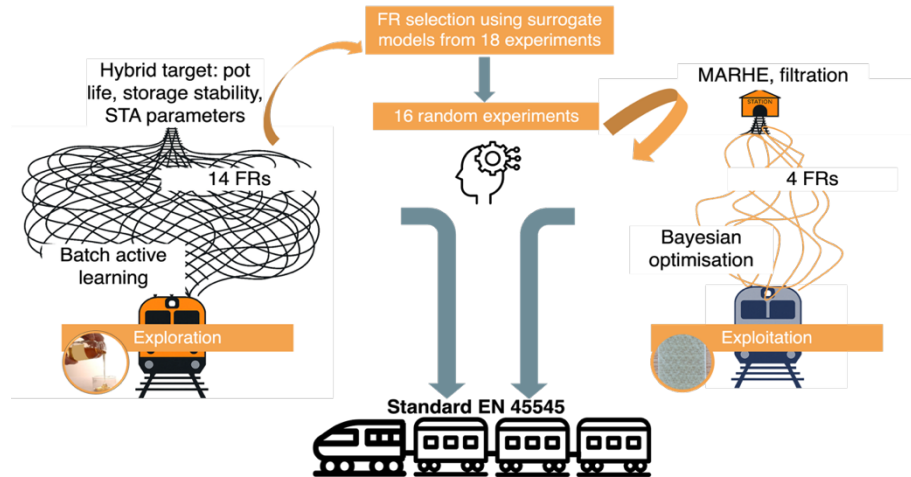


Figure 2: Overview of the methodology in the investigation

2.1 Resin preparation

A DGEBF resin system (Polyphlox 3721, Struktol GmbH) with an epoxy equivalent weight (EEW) of 235 g/Eq and a dynamic viscosity of 3000 mPa.s at room temperature containing 10% w/w of a RDP liquid FR was reacted with an IPDA-PACM hardener with an amine hydrogen equivalent weight (AHEW) of 47.6 g/Eq. DGEBF and IPDA were combined and dispersed by hand in the stoichiometric ratio of 100:22 based on the following equation:

$$\text{Resin} - \text{hardener ratio} = \left[\frac{100 \times (\text{AHEW}_{\text{IPDA-PACM}})}{(\text{EEW}_{\text{DGEBF}})} \right] \quad (1)$$

2.1 Flame retardants and glass fibers

Table 1 summarizes the various FR additives used within the study for feature selection screening. RDP is already mixed at a fixed proportion within the DGEBF resin. The halogenated flame retardant PropCP is considered for a comparison with the non-halogenated FRs and not intended for transfer to the composite. DOPOI is considered as an alternative liquid FR for transfer to the composite after feature importance determination from the existing 14 initial FRs. The particle size distribution of the 50% fraction of the FRs is reported as d_{50} in μm .

Chemical formulation	Commercial name	Supplier	Coating	d_{50} μm	Abbrev.	Form	P %
Resorcinol bis(diphenyl) phosphate	-	Struktol	-	-	RDP	Liquid	20
Ammonium polyphosphate	Polyphlox 3721 Comp. B		-	5	APPI	Solid	32
	FR CROS 484	Budenheim	-	20	APPII		

Ammonium polyphosphate	FR CROS 486	Budenheim	Silane	20	APPIII	Solid	32
	FR CROS C30		Melamine	7	APPV		
	FR CROS 584		-	18	APPV		
	FR CROS 585			18	APPVI		
Intumescent mixture with APP	FR CROS 638			12	APPVII		24
Melamine polyphosphate	FR CROS 513			8	MPP		13
Aluminium diethyl phosphinate	EXOLIT OP 935	Clariant		10	DEPALI		22
	EXOLIT OP 1230			20	DEPALI I		
Triethyl phosphate	Luvogard HF P70	Lehvoss		-	TEP	Liquid	18
2-Propanol, 1-chloro-, phosphate (3:1)	Fyrol PCF	ICL		-	ProCP		10
Zinc hydroxystannate	Flamtard H	William Blythe		5	ZHS	Solid	0
Inorganic silicate	Flamtard V100			5	Insi		
6H-Dibenz [c,e] [1,2] oxaphosphorin-6-propanoic acid, butyl ester, 6-oxide	Sacoflam 47201	Krems Chem		-	DOPOI	Liquid	20

Table 1: Database of FRs with varying physical forms, P content and chemical formulations for feature importance determination via B-AL trials

The variation in the fraction of the 14 FRs used for AL trials across 18 experiments is given in the heatmap in Figure 3. Aerospace textile satin woven glass fibers (GFs) (9111-FK144 from Valmiera glass) with an areal weight of $\sim 105 \text{ g/m}^2$ coated with a silane/volan finish were used for vacuum infusion.

3 METHODS

3.1 Active learning via batch experiments for FR selection

A library of fourteen phosphorus-based flame retardants (14 features) was initially gathered (Table 1), to evaluate their relative importance with respect to STA-derived thermal metrics, pot life, and storage stability to be determined. To minimize the number of physical trials, batch active learning employing a Gaussian process regression surrogate model was implemented: the first five formulations were sampled from a constrained virtual design space of 10,000 candidates via Gaussian sampling. The total FR loading was maintained $\leq 10 \text{ w/w}$, which including RDP was $\leq 20 \text{ w/w}$. In total, 18 experimental formulations, with 5 initial experiments and subsequently 3 per round were ultimately evaluated, and

Figure 3: FR loadings in w/w for the 18 experiments based on B-AL

A hybrid target function containing 12 sub-targets was defined as:

$$fFR1 = 0.25[(\text{Pot Life}) - (|T_{g0}|) + (\text{Positive parameters}) - (\text{Negative parameters})] \quad (2)$$

Here, the relevance of the positive and negative parameters is explained in Table 2. The STA parameters were each weighted based on the LASSO coefficients derived from the study by Binder et al.[15].

Influence on fire performance	Impact on pHRR	STA parameters
Positive influence on fire performance	Negative (increase in parameters implies a decrease in pHRR)	Initial sample mass, mass residue of the second peak in DSC, heat energy of the first curing peak in DSC , temperature at 50 % conversion of the first peak in TGA, temperature at 50 % conversion of the second peak, adjusted heat enthalpy of the first peak in DSC
Negative influence on fire performance	Positive (increase in parameters implies an increase in pHRR)	Total heat value in DSC, heat energy of the second curing peak in DSC , temperature of the second peak in DSC, heat flow of the first curing peak in DSC

Table 2: STA parameters positively and negatively influencing fire performance

3.2 Pot life

The influence of the FR additive combinations as additives for the DGEBF systems on pot life was measured using an Anton Paar Rheometer (MCR 301). The rheological properties of the resins were then evaluated at an isothermal sweep at 50 °C for 120 mins using a parallel plate (PP25) with a gap between parallel plates of 1 mm. The pot life in mins was determined as the time at which the resin reached a complex viscosity of 2000 mPa.s. A constant strain of 0.1% was used, with a maximum torque of 140 mN/m.

3.3 Storage stability

For the storage stability measurements, the DGEBF resin was not reacted with the hardener but directly combined with the FR additives and dispersed by hand. The resins were tested in a DSC1 from Mettler Toledo (USA) directly after preparation (Day 0). A sample mass of 5-10 mg, with a temperature range of -25 °C to 250 °C at a ramp rate of 10 K/min was used for each measurement. A constant flow of synthetic air at 50 mL/min was maintained. The samples were subsequently stored in a Memmert ULE 400 convection oven at 50 °C for 7 days after which a DSC measurement was performed. A benchmark of $\Delta T_{g0} \leq 10$ °C was used to determine the storage stability of the resins. ΔT_{g0} was calculated using the following equation:

$$\Delta T_{g0} = T_{g0}^{\text{Day7}} - T_{g0}^{\text{Day1}} \quad (2)$$

3.4 Simultaneous thermal analysis (STA)

For STA measurements, the FRs were dispersed into the resin-hardener mixture by hand mixing and cured overnight by film casting on a glass plate at 25 °C. Subsequently, the cured resin was comminuted in a cryomill at ~ 19,000 rpm using a 200 µm sieve. Samples with a mass of ~ 10 mg were tested simultaneously for DSC and TGA parameters in a Netzsch STA 449 F3 Jupiter using a temperature range from 25 °C to 800 °C with a ramp rate of 7.5 K/min. The various parameters mentioned in section 3.1 were then extracted.

3.5 Vacuum infusion and cone calorimetry

The resin formulations were preheated to 50 °C and infused into ~ 32 layers of the GFs stacked by hand under vacuum of 0 mbar. The infused GFs, with a target fiber volume content of 50 ± 5 % were cured overnight at room temperature and subsequently cut to size (100 mm x 100 mm x 3 mm) for cone calorimeter tests. The top and bottom faces of the samples were labelled. The samples were tested in a horizontal orientation – with one sample with the top face exposed to the cone heater, and with another with the bottom face exposed, under an external heat flux of 35 kWm^{-2} and 25 mm from the cone heater following ISO 5660-1 norms. The MARHE values and the standard deviation between the top and bottom face exposures from the cone tests was calculated to determine the presence of filtration effects.

4 RESULTS AND DISCUSSION

4.1 Pot life and storage stability

Figure 4 (a) reveals that the pot life values for all 18 formulations converge within a narrow range (± 10 min), suggesting a consistent impact of FR incorporation on resin rheology. No anomalous gelation behavior was observed, indicating thermal stability of the curing system under the evaluated formulations. Despite the high dimensionality of the features and targets within the AL experiments, a gradual increase in the value of the hybrid function is visible in Figure 4 (b).

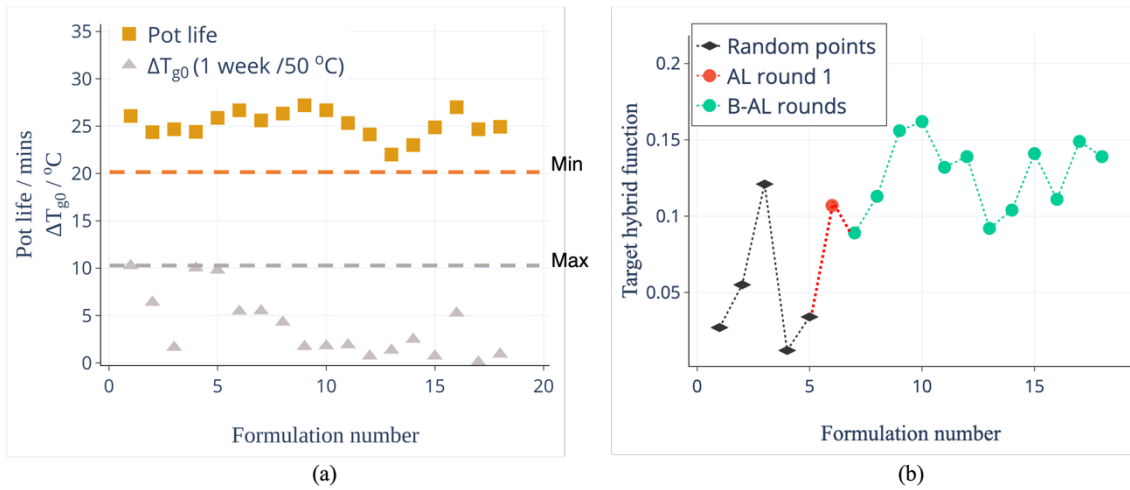


Figure 4: Variation of pot life and ΔT_{g0} for the resin formulations (a) and progress in the hybrid target across the 18 experiments (b) – Random, AL points

4.2 Training surrogate models for feature selection

The 14 input features (FR compositions) were scaled using None, MinMax, Standardisation and Robust scalers and either unpre-processed or pre-processed with techniques such as polynomial features of degree 2 (Poly2), first 5 principal components (PCA5) etc. The 12 sub-targets (see equation 2) from the 18 experiments via B-AL, the data were used as individual targets. The data from each sub-target was split into training test sets, to train 7 surrogate models – linear, random forest regression, gradient boosting regression (GBR), support vector regression (SVR), k-nearest neighbour (KNN), decision tree

(DT) and LASSO model. A five-fold cross-validation was employed for each surrogate model pipeline, ensuring that each of the 18 data points was held out once. The best pipelines of these 140 pipelines determined by those which predicted individual parameters with the highest cross validation (CV) R^2 values, lowest CV mean square error (MSE) and lowest CV mean absolute error (MAE) (Table 2) relative to the data, were selected to determine feature importance using permutation importance.

Target property (highlighted in table 2)	Pre- processing	Scaler	Surrogate model	CV R^2	CV MSE	CV MAE
Heat energy of the first curing peak	PCA	Standard	Linear	0.6320	0.0032	0.04805
Adjusted heat enthalpy of the first curing peak	PCA	None	Linear	0.5883	0.0004	0.01656
Heat energy of the second curing peak	Poly2	Robust	LASSO	0.4869	0.4298	0.5446

Table 3: Best pipelines based on 140 combinations of data preprocessing, scaling and training of 7 surrogate models

Thereby, the top ranked FRs based on feature importance were silane coated APP (APPIII), melamine coated APP (APPIV), proCP and TEP. Since proCP is halogenated, it was eliminated at this stage based on REACH regulations. While this 18-point AL dataset enabled proof-of-concept surrogate modelling, the moderate R^2 values indicate that additional rounds are required to fully capture nonlinear FR interactions. Future work will integrate high-throughput thermal screening to expand the design space. The high permutation importance of APPIII and APPIV is promising due to their known condensed-phase char-promoting behavior, while TEPs gas-phase radical-quenching action supports its intermediate ranking. DOPOI was selected as an additional liquid FR for 16 randomised trials in vacuum infusion. For this purpose, a second constrained virtual design space of 10,000 candidates via Gaussian sampling was generated with these 4 selected FRs, and 16 randomised trials were generated for vacuum infusion experiments and subsequent cone calorimeter tests.

4.3 Fire performance and filtration effects of the composites (under construction)

The pareto plot between TSP and MARHE of the randomized samples are shown in Figure 5. Sample number 14 was invalidated due to technical errors during cone calorimeter operation and showed the highest error in MARHE ($\pm 85 \text{ kW/m}^2$). Sample 13 containing a combination of 9% w/w of TEP with 1% w/w of APP simultaneously showed one of the lowest MARHE values and filtration effects. Although Sample 5, containing 5% w/w APPIII, 2% APPIV and 2% TEP showed the lowest MARHE values of all valid samples with significant intumescent behavior during fire testing, the difference from the top face (MARHE $\sim 70 \text{ kW/m}^2$) and bottom face was $\sim 30\%$. Thus, it can be hypothesized that although a potential synergistic effect is present in sample 5 with primarily condensed phase action from the APP leading to a HL-2 classification when the top face is exposed to a fire, the higher solid content present in the resin causes a gradient effect and reduces FR penetration and thus, fire performance towards the bottom of the composite. Compared to the composite without FRs, all samples containing at least 2% w/w of some combination of FRs improved the MARHE values, with the best formulation (sample 5) producing a $\sim 30\%$ reduction in MARHE.

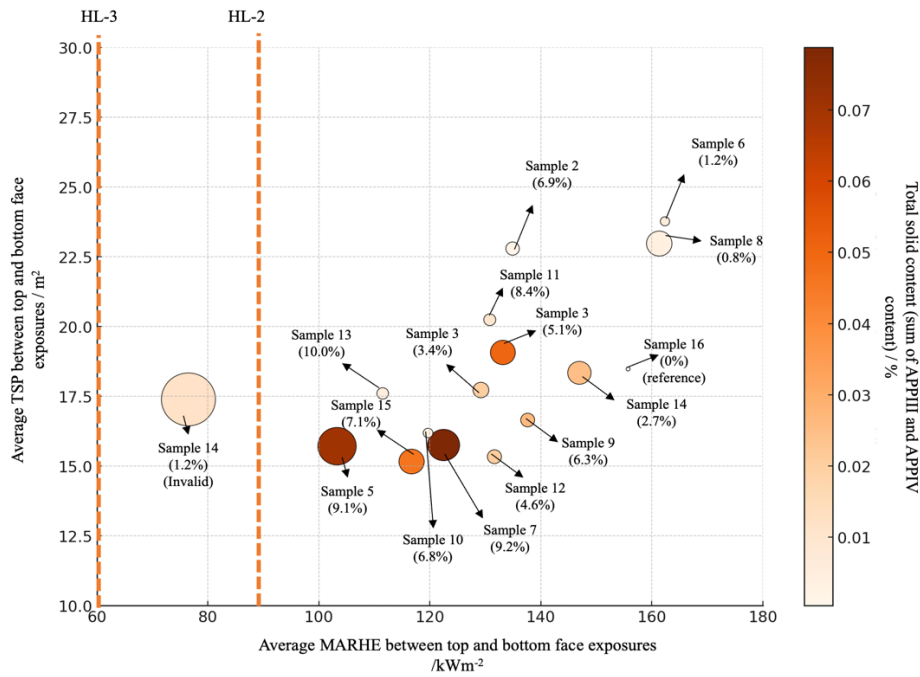


Figure 5: Pareto plot between average TSP and MARHE for the 16 samples showing solid content and filtration effects. Larger data points represent higher uncertainty in MARHE indicating higher filtration effects. All FR percentages are in addition to 10% w/w RDP content in the resin

5 CONCLUSION AND OUTLOOK

In this study, a systematic, ML-guided approach was used to identify and optimize P-based FRs as additives for EP GFRCs. From an initial pool of 14 candidate FRs, a B-AL framework using Gaussian-process surrogate models were used to efficiently screen formulations. By leveraging simultaneous thermal analysis (STA) parameters as rapid proxies for cone-calorimetry performance, we reduced the experimental burden completing 18 guided trials while still capturing feature importance. Coated APP formulations dominated the feature ranking, followed by a halogenated liquid FR and a TEP liquid FR.

16 randomized vacuum-infused composite specimens containing 4 selected FRs were fabricated and characterized via cone calorimetry under both top- and bottom-face exposures. Using only random trials, the MARHE values could be reduced by $\sim 30\%$, close to the HL-2 standard for railway applications. Although successful as a proof of concept, expanding the AL dataset will help capture nonlinear synergistic effects among multiple FRs. Validation under shear rates representative of injection molding ($\approx 10 \text{ s}^{-1}$) and scale-up trials in RTM will ensure further industrial applicability. Further improvement of the FR content in the resin is also required, potentially via Bayesian optimization rounds to balance filtration effects with MARHE values until a HL-2 classification is possible.

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